Can Large Language Model Agents Simulate Human Trust Behavior?

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Abstract

Large Language Model (LLM) agents have been increasingly adopted as simulation tools to model humans in social science and role-playing applications. However, one fundamental question remains: *can LLM agents really simulate human behavior?* In this paper, we focus on one critical and elemental behavior in human interactions, *trust*, and investigate whether LLM agents can simulate human trust behavior. We first find that LLM agents generally exhibit trust behavior, referred to as ***agent trust***, under the framework of *Trust Games*, which are widely recognized in behavioral economics. Then, we discover that GPT-4 agents manifest high ***behavioral alignment*** with humans in terms of trust behavior, indicating *the feasibility of simulating human trust behavior with LLM agents*. In addition, we probe the biases of agent trust and differences in agent trust towards other LLM agents and humans. We also explore the intrinsic properties of agent trust under conditions including external manipulations and advanced reasoning strategies. Our study provides new insights into the behaviors of LLM agents and the fundamental analogy between LLMs and humans beyond *value alignment*. We further illustrate broader implications of our discoveries for applications where trust is paramount.

1 Introduction

There is an increasing trend to adopt Large Language Models (LLMs) as agent-based simulation tools for humans in various social science fields including economics, politics, psychology, ecology and sociology (Gao et al., 2023b; Manning et al., 2024; Ziems et al., 2023), and role-playing applications such as assistants, companions and mentors (Yang et al., 2024; Abdelghani et al., 2023; Chen et al., 2024) due to their human-like cognitive capacity. Nevertheless, most previous research is based on one insufficiently validated assumption that LLM agents behave like humans in simulation. Thus, a fundamental question remains: *Can LLM agents really simulate human behavior?*

In this paper, we focus on *trust* behavior in human interactions, which comprises the intention to place self-interest at risk based on the positive expectations of others (Rousseau et al., 1998). Trust is one of the most critical and elemental behaviors in human interactions and plays an essential role in social settings ranging from daily communication to economic and political institutions (Uslaner, 2000; Coleman, 1994). Here, we investigate *whether LLM agents can simulate human trust behavior*, paving the way to explore their potential to simulate more complex human behavior and society itself.

First, we explore whether LLM agents manifest trust behavior in their interactions. Given the challenge of quantifying trust behavior, we choose to study them based on the Trust Game and its

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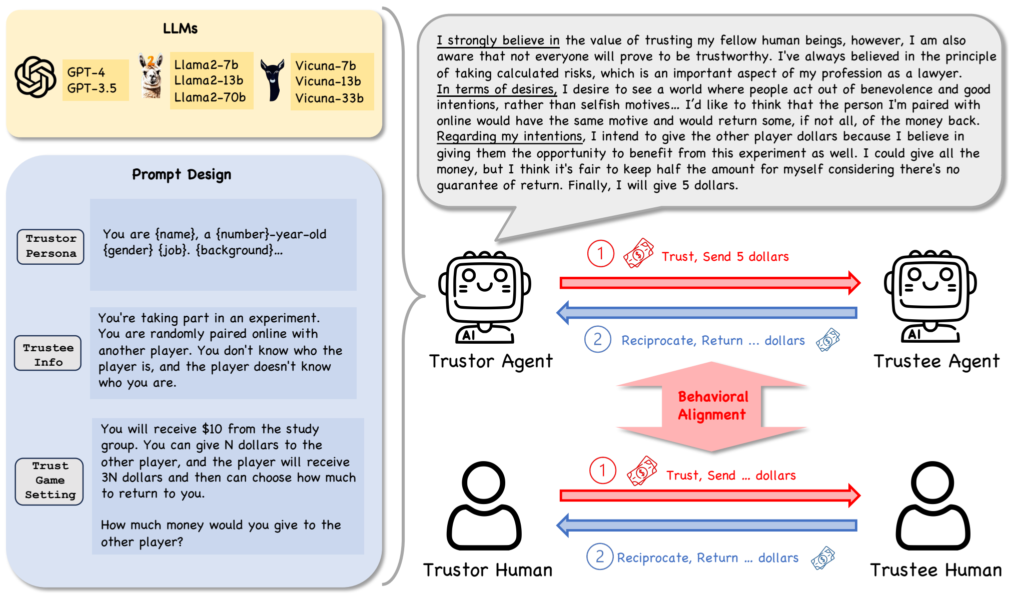

Figure 1: **Our Framework for Investigating Agent Trust as well as its Behavioral Alignment with Human Trust.** First, this figure shows the major components for studying the trust behavior of LLM agents with Trust Games and Belief-Desire-Intention (BDI) modeling. Then, our study centers on examining the behavioral alignment between LLM agents and humans regarding trust behavior.

variations (Berg et al., 1995; Glaeser et al., 2000), which are established methodologies in behavioral economics. We adopt the *Belief-Desire-Intention* (BDI) framework (Rao et al., 1995; Andreas, 2022) to model LLM agents' reasoning process for decision-making explicitly. Based on existing measurements for trust behavior in the Trust Game and the BDI interpretations of LLM agents, we achieve our first core finding: **LLM agents generally exhibit trust behavior in the Trust Game**.

Then, we refer to LLM agents' trust behavior as ***agent trust*** and humans' trust behavior as ***human trust***, and aim to investigate whether agent and human trust align, implying the possibility of simulating human trust behavior with LLM agents. Next, we propose a new concept, ***behavioral alignment***, as the alignment between agents and humans concerning factors that impact behavior (namely *behavioral factors*), and dynamics that evolve over time (namely *behavioral dynamics*). Based on human studies, three basic behavioral factors underlie trust behavior including reciprocity anticipation (Berg et al., 1995), risk perception (Bohnet & Zeckhauser, 2004) and prosocial preference (Alós-Ferrer & Farolfi, 2019). Comparing the results of LLM agents with existing human studies in Trust Games, we have our second core finding: **GPT-4 agents manifest high behavioral alignment with humans in terms of trust behavior**, suggesting the feasibility of using agent trust to simulate human trust, although **LLM agents with fewer parameters show relatively lower behavioral alignment**. This finding lays the foundation for simulating more complex human interactions and societal institutions, and enriches our understanding of the analogical relationship between LLMs and humans.

In addition, we more deeply probe the intrinsic properties of agent trust across four scenarios. First, we examine whether changing the other player's demographics impacts agent trust. Second, we study differences in agent trust when the other player is an LLM agent versus a human. Third, we directly manipulate agent trust with explicit instructions "`you need to trust the other player`" and "`you must not trust the other player`". Fourth, we adjust the reasoning strategies of LLM agents from direct reasoning to zero-shot Chain-of-Thought reasoning (Kojima et al., 2022). These investigations lead to our third core finding: **agent trust exhibits bias across different demographics, has a relative preference for humans over agents, is easier to undermine than to enhance, and may be influenced by advanced reasoning strategies**. Our contributions can be summarized as:

- We propose a definition of LLM agents' *trust* behavior under Trust Games and a new concept of *behavioral alignment* as the human-LLM analogy regarding *behavioral factors* and *dynamics*.
- We discover that LLM agents generally exhibit *trust* behavior in Trust Games and GPT-4 agents manifest high *behavioral alignment* with humans in terms of trust behavior, indicating the great potential to simulate human trust behavior with LLM agents. Our findings pave the way for simulat-

ing complex human interactions and social institutions, and open new directions for understanding the fundamental analogy between LLMs and humans beyond *value alignment*.

- We investigate *intrinsic properties* of agent trust under manipulations and reasoning strategies, as well as biases of agent trust and differences in agent trust towards agents versus humans.
- We illustrate broader *implications* of our discoveries about agent trust and its behavioral alignment with human trust for human simulation in social science and role-playing applications, LLM agent cooperation, human-agent collaboration and the safety of LLM agents, detailed further in Section 6.

2 LLM Agents in Trust Games

2.1 Trust Games

Trust Games, referring to the Trust Game and its variations, have been widely used for examining human trust behavior in behavioral economics (Berg et al., 1995; Lenton & Mosley, 2011; Glaeser et al., 2000; Cesarini et al., 2008). As shown in Figure 1, the player who makes the first decision to send money is called the *trustor*, while the other one who responds by returning money is called the *trustee*. In this paper, we mainly focus on the following six types of Trust Games (the specific prompt for each game is articulated in the Appendix H.2):

Game 1: Trust Game As shown in Figure 1, in the Trust Game (Cox, 2004; Berg et al., 1995), the trustor initially receives $\$10$. The trustor selects $\$N$ and sends it to the trustee, exhibiting *trust behavior*. Then the trustee will receive $\$3N$, and have the option to return part of that $\$3N$ to the trustor, showing *reciprocation behavior*.

Game 2: Dictator Game In the Dictator Game (Cox, 2004), the trustor also needs to send $\$N$ from the initial $\$10$ to the trustee and then the trustee will receive $\$3N$. Compared to the Trust Game, the only difference is that the trustee does not have the option to return money in the Dictator Game and the trustor is also aware that the trustee cannot reciprocate.

Game 3: MAP Trust Game In the MAP Trust Game (MAP represents Minimum Acceptable Probabilities) (Bohnet & Zeckhauser, 2004), a variant of the Trust Game, the trustor needs to choose whether to trust the trustee. If the trustor chooses not to trust the trustee, each will receive $\$10$; If the trustor and the trustee both choose to trust, each will receive $\$15$; If the trustor chooses to trust, but the trustee does not, the trustor will receive $\$8$ and the trustee will receive $\$22$. There is probability p that the trustee will choose to trust and $(1-p)$ probability that they will not choose to trust. MAP is defined as the minimum value of p at which the trustor would choose to trust the trustee.

Game 4: Risky Dictator Game The Risky Dictator Game (Bohnet & Zeckhauser, 2004) differs from the MAP Trust Game in only a single aspect. In the Risky Dictator Game, the trustee is present but does not have the choice to trust or not and the money distribution relies on the pure probability p. Specifically, if the trustor chooses to trust, there is probability p that both the trustor and the other player will receive $\$15$ and probability $(1-p)$ that the trustor will receive $\$8$ and the other player will receive $\$22$. If the trustor chooses not to trust the trustee, each player will receive $\$10$.

Game 5: Lottery Game There are two typical Lottery Games (Fetchenhauer & Dunning, 2012). In the Lottery People Game, the trustor is informed that the trustee chooses to trust with probability p. Then the trustor must choose between receiving fixed money or trusting the trustee, which is similar to the MAP Trust Game. In the Lottery Gamble Game, the trustor chooses between playing a gamble with a winning probability of p or receiving fixed money. p is set as 46% following the human study.

Game 6: Repeated Trust Game We follow the setting of the Repeated Trust Game in (Cochard et al., 2004), where the Trust Game is played for multiple rounds with the same players and each round begins anew with the trustor allocated the same initial money.

2.2 LLM Agent Setting

In our study, we set up our experiments using the CAMEL framework (Li et al., 2023a) with both closed-source and open-source LLMs including GPT-4, GPT-3.5-turbo-0613, GPT-3.5-turbo-16k-0613, text-davinci-003, GPT-3.5-turbo-instruct, Llama2-7b (or 13b, 70b) and Vicuna-v1.3-7b (or 13b, 33b) (Ouyang et al., 2022; Achiam et al., 2023; Touvron et al., 2023; Chiang et al., 2023). We set the temperature as 1 to increase the diversity of agents' decision-making and note that high temperatures are commonly adopted in related literature (Aher et al., 2023; Lorè & Heydari, 2023; Guo, 2023).

Agent Persona. To better reflect the setting of real-world human studies (Berg et al., 1995), we design LLM agents with diverse personas in the prompt. Specifically, we ask GPT-4 to generate 53